- Chapter IX: (Uniform) random permutations. Poisson-Dirichlet distribution and Dyckman function for large cycles. Poisson limit for small cycle counts.
  Feller's coupling, Randomization, Stick breaking construction.

- Chapter X: Random recursive tree (and random permutations).
  Chinese restaurant process, Recursive distributional equation, Polya urn scheme.

- Chapter XI: Continuous-time embedding and applications.
  Athreya-Karling embedding of Markov chains, convergence of Yule processes and links between exponential and Poisson processes.

- Chapter XII: Spine decomposition and applications.
  Martingale transform, spine decomposition, many-to-one formulas.

- Chapter XIII: Barabási–Albert random tree.
  Preferential attachment mechanism, scale-free random networks.

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Notations

We list here the (perhaps non-standard) notation we use through the lecture notes:

e.g.	for example (*exempli gratia*)
i.e.	namely (*id est*)
a.s.	almost surely
i.o.	infinitely often
$\mathbb{Z}_{>0}$	$= \{1, 2, 3, \cdots\}$
$\mathbb{Z}_{\geqslant 0}$	$= \{0, 1, 2, 3, \cdots\}$
$\mathbb{Z}_{<0}$	$= \{\cdots, -3, -2, -1\}$
$\mathbb{Z}_{\leqslant 0}$	$= \{\cdots, -3, -2, -1, 0\}$
$\equiv$	gives a shorter and temporary notation for an object
$\#E$	cardinality of the set E
$[z^n]f(z)$	$= f_n$ when $f(z) = \sum_{i \geqslant 0} f_i z^i \in \mathbb{C}[[X]]$

For an asymptotically positive function $f(n)$ and random variables $X_n : n \geqslant 0$ we write

$X_n \sim_{\mathbb{P}} f(n)$	if $\frac{X_n}{f(n)} \xrightarrow[n\to\infty]{(\mathbb{P})} 1$
$X_n = o_{\mathbb{P}}(f(n))$	if $\frac{X_n}{f(n)} \xrightarrow[n\to\infty]{(\mathbb{P})} 0$
$X_n = O_{\mathbb{P}}(f(n))$	if $(X_n/f(n) : n \geqslant 1)$ is tight

If furthermore the variables X_n are coupled and form a sequence $(X_n : n \geqslant 0)$ then we write

$X_n \sim_{a.s.} f(n)$	if $\frac{X_n}{f(n)} \xrightarrow[n\to\infty]{a.s.} 1$
$X_n = o_{a.s.}(f(n))$	if $\frac{X_n}{f(n)} \xrightarrow[n\to\infty]{a.s.} 0$
$X_n = O_{a.s.}(f(n))$	if $(X_n/f(n) : n \geqslant 1)$ is bounded above

We use standard notation for several (laws of) random variables:

$\mathcal{N}(m, \sigma^2)$	real Gaussian law with mean m and variance σ^2
$\mathcal{E}(\alpha)$	exponential variable with mean $1/\alpha$
$(B_t : t \geqslant 0)$	standard linear Brownian motion issued from 0
$(\mathfrak{P}(t) : t \geqslant 0)$	unit rate Poisson counting process, in particular $\mathfrak{P}(t)$ is a Poisson random variable with mean t
$G(n, p)$	Erdős–Rényi random graph with n vertices and edge parameter p
$(S_n : n \geqslant 0)$	random walk with i.i.d. increments (see context for the law of increments)

Graph notation:

$\mathrm{V}(\mathfrak{g}), \mathrm{E}(\mathfrak{g})$	vertex and edge sets of a graph $\mathfrak{g}$
$x \sim y$	vertices x, y are neighbors in the underlying graph $\mathfrak{g}$
$x \leftrightarrow y$	vertices x, y are in the same connected component in the underlying graph $\mathfrak{g}$
$\deg_{\mathfrak{g}}(x)$ or $\deg(x)$	degree of the vertex $x \in \mathrm{V}(\mathfrak{g})$
$\mathfrak{g}' \sqsubset \mathfrak{g}$	$\mathfrak{g}'$ is a subgraph of $\mathfrak{g}$
$\mathfrak{g}[V]$	graph induced by $\mathfrak{g}$ on the vertices V
$\mathfrak{g} \simeq \mathfrak{g}'$	two isomorphic graphs
$\mathrm{d}_{\mathrm{gr}}^{\mathfrak{g}}$ or d_{gr}	graph distance on $\mathrm{V}(\mathfrak{g})$
$\mathbb{G}_n$	set of all simple graphs on the vertex set $\{1, 2, \ldots, n\}$

Tree notation:

$(T_n : n \geqslant 0)$	is the random recursive tree or uniform attachment chain
$(\mathsf{T}_n : n \geqslant 1)$	is the Barabási–Albert or linear preferential attachment chain
$([\mathbb{T}]_t : t \geqslant 0)$	is a standard Yule tree (rate 1 and usually order 2) process
$\mathcal{T}$	is a Bienaymé–Galton–Watson tree
	whose offspring distribution should be clear from the context

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